Quality Control Layer 1: Research & Factual Accuracy Audit

Lesson: HTTP Requests and Responses | **Audit Reference:** AUDIT-FACT-0001

Overall Audit Status: NEEDS IMPROVEMENT **Factual Accuracy Score:** 90.0% **Issues Detected:** 3

1. Executive Accuracy Assessment

The lesson accurately conveys core HTTP semantics (GET/POST safety and idempotency, 2xx/4xx/5xx status classes) consistent with RFC 9110 and MDN Web Docs. However, it contains citation inaccuracies: status codes 422 and 429 are not defined in RFC 9110 (they originate in RFC 4918 and RFC 6585 respectively), and the lesson cites a book available only as metadata/preview as verification for specific technical content.

2. Factual Claims Analysis & Corrections (3 items)

ID / Severity	Affected Section & Claim	Audit Finding & Supporting Evidence	Recommended Correction
FACT-001 MEDIUM	**Section:** Section 4 - Status Codes for Debugging (4xx Client Error) *"422 Unprocessable Entity: The request was well-formed but semantically invalid (e.g., validation error). 429 Too Many Requests: Rate limiting has been applied."*	RFC 9110 (HTTP Semantics) does not define status codes 422 or 429. Code 422 is defined in RFC 4918 (WebDAV) and 429 in RFC 6585. While MDN documents both, listing them under a section cited to RFC 9110 implies they are part of the core HTTP/1.1 semantics specification, which is inaccurate. **Evidence:** RFC 9110 Section 15 does not list 422 or 429; RFC 4918 Section 11.2 defines 422; RFC 6585 defines 429.	**Fix:** Add a clarifying note that 422 and 429 are extension status codes (RFC 4918 and RFC 6585 respectively), or remove them from the RFC 9110-cited list and cite the appropriate extension RFCs.
FACT-002 LOW	**Section:** Sections 4 and 6 (Debugging Mindset; Real World Applications) *"Inline citations to [HTTP: The Definitive Guide] supporting technical claims about HTTP status codes and debugging practices."*	Source [3] is accessible only as Google Books metadata/preview; the full text is not available for verification. Citing it as an authority for specific lesson content cannot be substantiated from the provided research sources. **Evidence:** Source metadata for [3] indicates 'Accessible: Metadata/Preview only'.	**Fix:** Remove [HTTP: The Definitive Guide] from inline technical citations or re-label it as supplementary background reading only, not a verification source for specific claims.

FACT-003 LOW	**Section:** Section 4 - GET Requests *"Because GET is idempotent, browsers may cache responses, and clients can safely retry failed GET requests."*	RFC 9110 defines GET as safe and idempotent, but idempotency does not guarantee that retrying a failed request is safe in all network failure scenarios (e.g., if the server processed the request but the response was lost before reaching the client). This is a common simplification but could mislead learners about distributed systems safety. **Evidence:** RFC 9110 Section 9.2.1 (Safe Methods) and Section 9.2.2 (Idempotent Methods).	**Fix:** Because GET is safe and idempotent, browsers may cache responses, and clients can generally retry failed GET requests, though network-level failures require care.

3. Verification Scope & Limitations

Source [3] (HTTP: The Definitive Guide) is available only as metadata/preview, so full-text verification of its claims is impossible. Source [4] is a YouTube video; no transcript was provided, so specific verbal claims cannot be textually verified. Only Sources [1] (RFC 9110) and [2] (MDN Web Docs) were fully accessible for verification.

4. Conclusion & Certification

The lesson is factually accurate on core HTTP method semantics and status code classes per RFC 9110 and MDN, but requires minor corrections to citation attribution for extension status codes (422, 429) and an unverifiable book preview source.